

A harmonised point-extraction and comparison tool for heterogeneous paleoclimate datasets

Geoinformatics Project Report

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Abstract

1 Intro

- Paleoclimate models are useful for:
 - Reconstructing Earth's climate before direct measurements existed
 - Understanding how the climate responds to forcings like solar output, volcanic eruptions, orbital cycles, and CO₂
 - Testing and validating modern climate models against known past states
 - Establishing the natural variability range of climate, separate from human influence
 - Constraining climate sensitivity (temperature change per CO₂ doubling)
 - Studying planetary climate evolution — Mars, Venus, and other terrestrial bodies
 - Using past warm periods (e.g. Pliocene) as analogues for near-future climate projections
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- there are many different paleoclimate models.
 - comparing them is important to detect outliers, irregularities or strong differences eg. to help improve the models

2 The Data

- can be downloaded via `data_downloader.ipynb`

2.1 Trace21k

- Simulation of the Transient Climate of the Last 21,000 Years - Chosen Version: TraCE Main Land Decadal Average
- developed by: see source
- year of development: see source
- how where they made: fully-coupled, non-accelerated atmosphere-ocean-sea ice-land surface CCSM3 simulation
- what was the goal:
- nr. files: 2
- size (gb): 449,2 MB
- type: netcdf
- used variables: TSA, TSOI

- how to get: (via python script in my data_downloader.ipynb notebook)
- other info:
- extend time: 22,000 BCE to 1990 CE
- resolution time: decadal averages
- extend space: global land
- resolution space: T31_gx3
- source: Otto-Bliesner and Rosenbloom 2021

2.2 CHELSA-TraCE21k

- Chosen Version: CHELSA-TraCE21k-centennial
- developed by: see source
- year of development: see source
- how where they made: generated with the CHELSA-TraCE21k downscaling model.
- what was the goal:
- nr. files:
- size (gb):
- type: geotiff -> need to be converted to netcdf
- used variables: tasmin, tasmax -> $\text{tasmin} + \text{tasmax}/2$ to get mean
- how to get: (via python script in my data_downloader.ipynb notebook)
- other info:
- extend time: Start Time 21k BP, End Time 0 BP
- resolution time: monthly averages, 100 year timesteps
- extend space: global
- resolution space: kilometer scale
- source: Karger et al. 2023, Karger et al. 2020
- because of big filesize i download tasmax tiffs, convert to netcdf, compute decadal mean, delete the tifs and then do the same for tasmin

2.3 Beyer 2021 (v4)

- Chosen Version: 12293345 version 4 (figshare, published 24 June 2021)
- developed by: Robert M. Beyer, Mario Krapp & Andrea Manica (Department of Zoology, University of Cambridge)
- year of development: original paper published 14 July 2020 (received 03 July 2019); dataset v4 published 24 June 2021; addendum paper published 29 September 2021
- how were they made: Delta-method-based downscaling and bias correction, combining

medium-resolution HadCM3 transient simulations with high-resolution HadAM3H snapshots and present-day observational data

- what was the goal: Fill the gap between high-temporal-resolution but low-spatial-resolution palaeoclimate simulations and high-spatial-resolution snapshot-only datasets
- nr. files:
- size (GB):
- type: NetCDF (data) + R / Python / MATLAB scripts
- used variables:
- how to get: (via python script in my data_downloader.ipynb notebook)
- other info: v4 adds global monthly NPP relative to v3; documented in the 2021 addendum
- extend time: 120000 BP to pre-industrial modern era (0 BP)
- resolution time: 2000-yr steps from 120000–22000 BP; 1000-yr steps from 22000 BP to pre-industrial (72 snapshots total)
- extend space: global terrestrial (full 360° longitude; latitude coverage 720x 300 cells at 0.5°, i.e. 150° of latitude)
- resolution space: 0.5° equirectangular grid (~55 km at the equator)
- source: Beyer, Krapp, and Manica 2020, Beyer, Krapp, and Manica 2021a, Beyer, Krapp, and Manica 2021b

2.4 PalMod 2

- Chosen Version: PalMod2 MPI-M MPI-ESM1-2-CR Transient Simulations of the Last Deglaciation with prescribed ice sheets from GLAC-1D reconstructions (r1i1p1f1)
- developed by: Mikolajewicz, U.; Kapsch, M.-L.; Gayler, V.; Meccia, V. L.; Riddick, T.; Ziemen, F. A.; Schannwell, C. (Max Planck Institute for Meteorology / DKRZ)
- year of development: dataset DOI published 2023
- how where they made: fully-coupled transient simulation with the Max Planck Institute Earth System Model version 1.2 in coarse resolution (MPI-ESM1-2-CR), combining the ECHAM6.3 spectral atmosphere (T31, 31 vertical levels) - what was the goal: produce a transient, fully-coupled simulation of the Last Deglaciation with realistic, time-evolving ice-sheet, topography and land-sea-mask forcing as part of the PalMod2 project
- nr. files:
- size (gb):
- type: NetCDF
- used variables: tas (PMMXMCRTDGr111Amtasgn30201, near-surface air temperature, monthly) tsl: (PMMXMCRTDGr111Lmtslgn30201, soil temperature by layer, monthly)

- how to get: WDCC archive at DKRZ; download via jblob client after registration World Data Center for Climate and Deutsches Klimarechenzentrum 2023 (via python script in my data_downloader.ipynb notebook)
- other info:
- extend time: 25000 BP to 0 BP (= 1950 CE)
- resolution time: variable-dependent — monthly for the standard atmosphere/land/ocean/sea-ice tables (Amon, Lmon, Omon, SImon, LImon, Emon), decadal (dec, Odec) and fixed (fx, Ofx); GLAC-1D ice-sheet/topography forcing updated every 10 years
- extend space: global
- resolution space: atmosphere/land T31 ($\sim 3,75^\circ$), ocean MPIOM1.6 nominal $\sim 3^\circ$
- source: Mikolajewicz et al. 2023

3 Homogenization

3.1 Variables

- which do i choose, are they directly comparable?

3.2 Time

- time extend
- time resolution

3.3 Space

- grid sizes
- changes in north south
- different resolutions for different times

4 Point Extraction

- do i just take the grid cell the point is in?

5 Results)

- For one example Point
- differences between models

6 Future

- Auto iterate over points and compare
- correlate with eg. north south, climate zones, water vs land etc.

7 Conclusion

- hard to compare due to differences eg. resolutions
- comparison can help to find out where they differ the most -> conclusion to how to improve models?

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